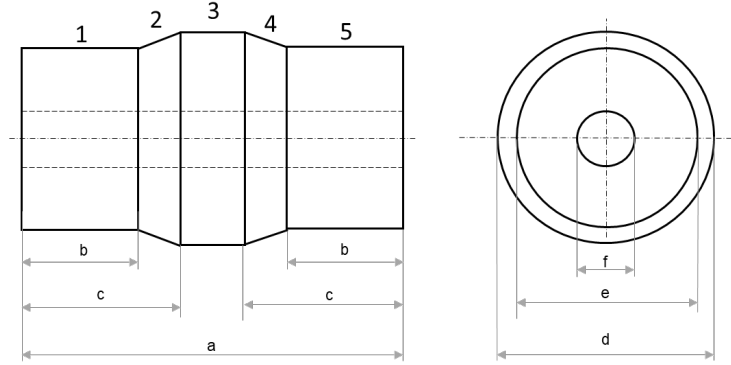


METAL CASTING

Problem 1 (06/11/2020)

The figure shows a casting to be fabricated by an expendable mould process. The casting will be made of AISI 304 steel (for simplicity, draft angles and fillets/radii are omitted). The casting is divided in 5 elementary geometries (numbered from 1 to 5).



- 1) Please, study the directional solidification of the metal. Dimensional data: $a = 340$ mm, $b = 115$ mm, $c = 145$ mm, $d = 165$ mm, $e = 145$ mm, $f = 60$ mm.
- 2) Considering: the casting axis as parting line, $A_s:A_r:A_g = 2:2:1$, $c = 0,6$, two gates with an area equal to 700 mm² each, a volume of the casting equal to 5 dm³, a top riser with a volume equal to 3 dm³, a cope height equal to 300 mm. Design a middle gating system, verifying if the mould filling time is lower than 10 s.

Solution

- 1) Please, study the directional solidification of the metal.

To study the directional solidification of the metal, we need to evaluate the ratio $M=V/A$ for each of the elementary geometries.

Geometry n.1 (equal to Geometry n.5):

$$V_1 = \frac{\pi}{4}(e^2 - f^2)b = \frac{\pi}{4}(145^2 - 60^2)115 = 1.573.840 \text{ mm}^3$$

$$A_1 = \pi(e + f)b + \frac{\pi}{4}(e^2 - f^2) = \pi(145 + 60)115 + \frac{\pi}{4}(145^2 - 60^2) = 87.749 \text{ mm}^2$$

$$M_1 = \frac{V_1}{A_1} = \frac{1.573.840}{87.749} = 17,94 \text{ mm} = M_5$$

Geometry n.2 (equal to Geometry n.4):

Being $L_2 = c - b = 30$ mm the length of the truncated cone,

$$V_2 = \frac{1}{12}\pi(d^2 + e^2 + de)L_2 - \frac{\pi}{4}(f^2)L_2 = \frac{1}{12}\pi(165^2 + 145^2 + 165 \cdot 145)30 - \frac{\pi}{4}(60^2)30 = 482.038 \text{ mm}^3$$

$$A_2 = \frac{\pi}{2}(d + e)\sqrt{L_2^2 + \left(\frac{d}{2} - \frac{e}{2}\right)^2} + \pi f L_2 = \frac{\pi}{2}(165 + 145)\sqrt{30^2 + \left(\frac{165}{2} - \frac{145}{2}\right)^2} + \pi 60 \cdot 30 = 21.054 \text{ mm}^2$$

$$M_2 = \frac{V_2}{A_2} = \frac{482.038}{21.054} = 22,90 \text{ mm} = M_4$$

Geometry n.3:

Being $L_3 = a - 2c = 50$ mm the length of cylinder,

$$V_3 = L_3 \frac{\pi}{4} (d^2 - f^2) = 50 \frac{\pi}{4} (165^2 - 60^2) = 927.752 \text{ mm}^3$$

$$A_3 = \pi(d + f)L_3 = \pi(165 + 60)50 = 35.343 \text{ mm}^2$$

$$M_3 = \frac{V_3}{A_3} = \frac{927.752}{35.343} = 26,25 \text{ mm}$$

So $M_3 > M_2 > M_1$ and $M_3 > M_4 > M_5$, with

$$\frac{M_2}{M_1} = \frac{M_4}{M_5} = \frac{22,90}{17,94} = 1,28 \in [1,1 \div 1,3]$$

$$\frac{M_3}{M_2} = \frac{M_3}{M_4} = \frac{26,25}{22,90} = 1,15 \in [1,1 \div 1,3]$$

Therefore, the solidification will start in geometries 1 and 5, and will end in geometry 3.

A riser should be positioned in contact with the geometry 3.

2) Design a middle gating system, verifying if the mould filling time is lower than 10 s.

In order to find the mould filling time, we need to know the area of the bottleneck and the velocity of the molten metal in this area. Being by design $A_s : A_r : A_g = 2 : 2 : 1$, the bottleneck is in the gates. The overall area of the gates is equal to:

$$A_g = 2 \cdot 700 = 1.400 \text{ mm}^2$$

In order to determine the velocity, being a middle gating system, we need to recall that:

$$v = c\sqrt{2gH_{av}}$$

where:

$$H_{av} = \left(\frac{1}{\frac{r_{drag}}{\sqrt{h_i}} + \frac{r_{cope}}{\frac{\sqrt{h_i} + \sqrt{h_f}}{2}}} \right)^2$$

$$V_{total} = V_{casting} + V_{riser} = 5 + 3 = 8 \text{ dm}^3$$

$$V_{cope} = \frac{V_{casting}}{2} + V_{riser} = 2,5 + 3 = 5,5 \text{ dm}^3 \rightarrow r_{cope} = \frac{V_{cope}}{V_{total}} = \frac{5,5}{8} = 0,6875 = 68,75\%$$

$$V_{drag} = \frac{V_{casting}}{2} = 2,5 \text{ dm}^3 \rightarrow r_{drag} = \frac{V_{drag}}{V_{total}} = \frac{2,5}{8} = 0,3125 = 31,25\%$$

Being $h_i = 300$ mm and $h_f = 0$ mm (we are using a top riser), we get:

$$H_{av} = \left(\frac{1}{\frac{0,3125}{\sqrt{300}} + \frac{0,6875}{\frac{\sqrt{300} + \sqrt{0}}{2}}} \right)^2 = 105,35 \text{ mm} = 0,10535 \text{ m}$$

Therefore:

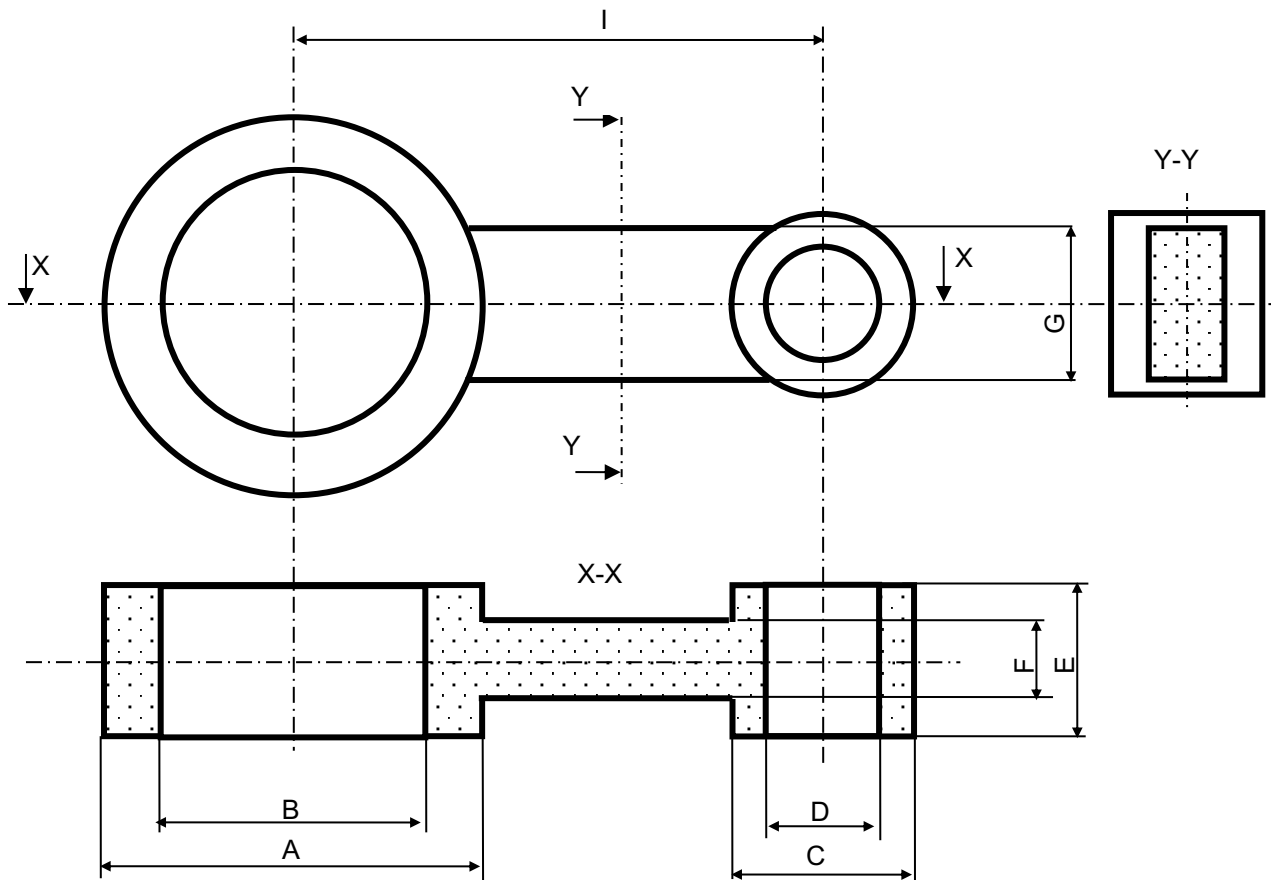
$$v = c\sqrt{2gH_{av}} = 0,6\sqrt{2 \cdot 9,81 \cdot 0,10535} = 0,863 \frac{\text{m}}{\text{s}} = 863 \frac{\text{mm}}{\text{s}}$$

and the mould filling time is:

$$T_{MF} = \frac{V_{total}}{A_g \cdot v} = \frac{8.000.000 \text{ mm}^3}{1.400 \text{ mm}^2 \cdot 863 \frac{\text{mm}}{\text{s}}} = 6,62 \text{ s} < 10 \text{ s}$$

The mould filling time satisfies the constraint.

Problem 2 (15/01/2021)



The figure shows a casting to be fabricated by an expendable mould process. The casting will be made of steel and will have the following final dimensions after machining: $A = 50$ mm, $B = 35$ mm, $C = 24$ mm, $D = 15$ mm, $E = 20$ mm, $F = 10$ mm, $G = 20$ mm, $I = 70$ mm.

- 1) Design the pattern and any cores respecting a machining allowance equal to 2 mm and a shrinkage allowance equal to 1,4% (for simplicity, draft angles and fillets/radii are omitted).
- 2) Considering the original dimensions, study the directional solidification of the metal. Is the directional solidification feasible? (Consider three elements: Element A is the cylinder of external diameter A; Element C is cylinder of external diameter C, Element B is the middle block to be considered tangent to the cylinders)
- 3) Considering: $A_s:A_r:A_g = 4:3:2$, $c = 0,5$, a volume of the casting equal to 36 cm^3 , a top riser with a volume equal to 15 cm^3 , a distance of the upper plane of the mould from the lower plane of the cavity equal to 200 mm. Design a bottom gating system so that the mould filling time is equal to 12 s.

Solution

- 1) Design the pattern and any cores respecting a machining allowance equal to 2 mm and a shrinkage allowance equal to 1,4% (for simplicity, draft angles and fillets/radii are omitted).

To find the pattern dimensions, we need to consider the material allowance needed for the following machining operations and the shrinkage allowance (1,4%). The result must be rounded to mm: upper integer for external dimensions, lower integer for internal dimensions. Therefore (for sake of completeness the computation is done for all the pattern dimensions):

$$A = (50 + 2 \cdot 2)(1 + 0,014) = 54 \cdot 1,014 = 54,756 = 55 \text{ mm}$$

$$B = (35 - 2 \cdot 2)(1 + 0,014) = 31 \cdot 1,014 = 31,434 = 31 \text{ mm}$$

$$C = (24 + 2 \cdot 2)(1 + 0,014) = 28 \cdot 1,014 = 28,392 = 29 \text{ mm}$$

$$D = (15 - 2 \cdot 2)(1 + 0,014) = 11 \cdot 1,014 = 11,154 = 11 \text{ mm}$$

$$E = (20 + 2 + 2)(1 + 0,014) = 24 \cdot 1,014 = 24,336 = 25 \text{ mm}$$

$$F = (10 + 2 + 2)(1 + 0,014) = 14 \cdot 1,014 = 14,196 = 15 \text{ mm}$$

$$G = (20 + 2 + 2)(1 + 0,014) = 24 \cdot 1,014 = 24,336 = 25 \text{ mm}$$

$$I = 70 \cdot (1 + 0,014) = 70 \cdot 1,014 = 70,98 = 71 \text{ mm}$$

2) Considering the original dimensions, study the directional solidification of the metal. Is the directional solidification feasible?

Element A:

$$V_A = \frac{\pi}{4}(A^2 - B^2) \cdot E = \frac{\pi}{4}(50^2 - 35^2) \cdot 20 = 20.028 \text{ mm}^3$$

$$S_A = 2 \cdot \frac{\pi}{4}(A^2 - B^2) + \pi(A + B) \cdot E - F \cdot G = 2 \cdot \frac{\pi}{4}(50^2 - 35^2) + \pi(50 + 35) \cdot 20 - 10 \cdot 20 = 7.143 \text{ mm}^2$$

$$M_A = \frac{V_A}{S_A} = \frac{20.028}{7.143} = 2,80 \text{ mm}$$

Element C:

$$V_C = \frac{\pi}{4}(C^2 - D^2) \cdot E = \frac{\pi}{4}(24^2 - 15^2) \cdot 20 = 5.513 \text{ mm}^3$$

$$S_C = 2 \cdot \frac{\pi}{4}(C^2 - D^2) + \pi(C + D) \cdot E - F \cdot G = 2 \cdot \frac{\pi}{4}(24^2 - 15^2) + \pi(24 + 15) \cdot 20 - 10 \cdot 20 = 2.802 \text{ mm}^2$$

$$M_C = \frac{V_C}{S_C} = \frac{5.513}{2.802} = 1,97 \text{ mm}$$

Element B:

$$L_B = I - \frac{A}{2} - \frac{C}{2} = 70 - \frac{50}{2} - \frac{24}{2} = 33 \text{ mm}$$

$$V_B = F \cdot G \cdot L_B = 10 \cdot 20 \cdot 33 = 6.600 \text{ mm}^3$$

$$S_B = 2 \cdot F \cdot L_B + 2 \cdot G \cdot L_B = 2L_B(F + G) = 2 \cdot 33(10 + 20) = 1.980 \text{ mm}^2$$

$$M_B = \frac{V_B}{S_B} = \frac{6.600}{1.980} = 3,33 \text{ mm}$$

Directional solidification:

The first element that will solidify is Element C, then Element A and at last Element B. Being Element B in between Elements A and C, the riser must be positioned on the Element B (last to solidify).

Comparing the V/A ratios, it results:

$$\frac{M_B}{M_A} = \frac{3,33}{2,80} = 1,19 \in [1,1 \div 1,3] \rightarrow \text{Element B will correctly feed Element A}$$

$$\frac{M_B}{M_C} = \frac{3,33}{1,97} = 1,69 \in [1,1 \div 1,3] \rightarrow \text{Element B will feed Element C, but the ratio is too high}$$

Therefore, a change is needed to have a feasible casting process.

3) Considering: the bottom plane of the casting as parting line, $A_s:A_r:A_g = 4:3:2$, $c = 0,5$, a volume of the casting equal to 36 cm^3 , a top riser with a volume equal to 15 cm^3 , a cope height equal to 200 mm. Design a bottom gating system so that the mould filling time is equal to 12 s.

In the problem, a bottom gating system is used to fill a mould with a top riser. Therefore:

$$h_i = 200 \text{ mm}$$

$$h_f = 0 \text{ mm}$$

$$H_{av} = \left(\frac{\sqrt{h_i} + \sqrt{h_f}}{2} \right)^2 = \left(\frac{\sqrt{200} + \sqrt{0}}{2} \right)^2 = 50 \text{ mm}$$

Therefore, the pouring velocity in the bottleneck cross section (area of the gates, A_g) will be

$$v = c \cdot \sqrt{2 \cdot g \cdot H_{av}} = 0,5 \cdot \sqrt{2 \cdot 9,81 \cdot \frac{50}{1000}} = 0,495 \frac{\text{m}}{\text{s}} = 495 \frac{\text{mm}}{\text{s}}$$

Being the mould filling time equal to $T_{MF} = \frac{V_{total}}{A_{bottleneck} \cdot v}$, we get:

$$V_{total} = V_{casting} + V_{risers} = 36 + 15 = 51 \text{ cm}^3 = 51.000 \text{ mm}^3$$

$$A_{bottleneck} = \frac{V_{total}}{T_{MF} \cdot v} = \frac{51.000}{12 \cdot 495} = 8,59 \text{ mm}^2$$

Therefore, the design of the gating system will be characterized by the following areas of the gates, runners and downsprue:

$$A_g = A_{bottleneck} = 8,59 \text{ mm}^2$$

$$A_r = \frac{3}{2} A_g = \frac{3}{2} \cdot 8,59 = 12,89 \text{ mm}^2$$

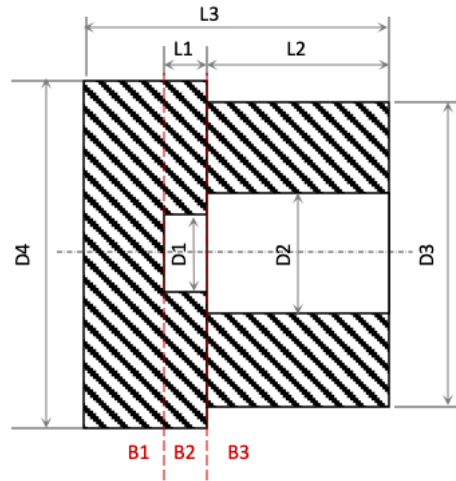
$$A_s = \frac{4}{2} A_g = 2 \cdot 8,59 = 17,18 \text{ mm}^2$$

Problem 3 (05/02/2021)

The figure shows an axisymmetric casting to be fabricated by an expendable mould process.

Data: $D_1 = 50$ mm, $D_2 = 80$ mm, $D_3 = 200$ mm, $D_4 = 230$ mm, $L_1 = 30$ mm, $L_2 = 120$ mm, $L_3 = 200$ mm.

- 1) Design the core respecting a machining allowance equal to 2 mm and a shrinkage allowance equal to 1% (for simplicity, draft angles and fillets/radii are omitted).
- 2) Considering the original dimensions and the elements as in the figure (B1, B2, and B3), study the directional solidification of the metal. On top of which element should a riser be positioned?
- 3) Considering: the axis of the casting as parting line, a volume of the casting cavity equal to $6,5 \text{ dm}^3$, a riser with a volume equal to 3 dm^3 , a middle gating system with H_{av} equal to 100 mm and $c = 0,6$. Determine the area of the bottleneck so that the mould filling time is equal to 10 s.



Solution

- 1) Design the core respecting a machining allowance equal to 2 mm and a shrinkage allowance equal to 1% (for simplicity, draft angles and fillets/radii are omitted).

To find the core dimensions, we need to consider the material allowance needed for the following machining operations and the shrinkage allowance (1,4%). The result must be rounded to mm. Therefore, the core dimensions are:

$$D_1 = (50 - 2 \cdot 2)(1 + 0,01) = 46 \cdot 1,01 = 46,46 \rightarrow D_1 = 46 \text{ mm}$$

$$D_2 = (80 - 2 \cdot 2)(1 + 0,01) = 76 \cdot 1,01 = 76,76 \rightarrow D_2 = 76 \text{ mm}$$

$$L_1 = (30 - 2 + 2)(1 + 0,01) = 30 \cdot 1,01 = 30,3 \rightarrow L_1 = 30 \text{ mm}$$

$$L_2 = (120 - 2 + 2)(1 + 0,01) = 120 \cdot 1,01 = 121,2 \rightarrow L_2 = 121 \text{ mm}$$

- 2) Considering the original dimensions and the elements as in the figure (B1, B2, and B3), study the directional solidification of the metal. On top of which element should a riser be positioned?

Element B1:

$$V_{B1} = \frac{\pi}{4} D_4^2 \cdot (L_3 - L_1 - L_2) = \frac{\pi}{4} 230^2 \cdot (200 - 30 - 120) = \frac{\pi}{4} 230^2 \cdot 50 = 2.077.378 \text{ mm}^3$$

$$S_{B1} = \frac{\pi}{4} D_4^2 + \frac{\pi}{4} D_1^2 + \pi D_4 \cdot (L_3 - L_1 - L_2) = \frac{\pi}{4} 230^2 + \frac{\pi}{4} 50^2 + \pi \cdot 230 \cdot 50 = 79.639 \text{ mm}^2$$

$$M_{B1} = \frac{V_{B1}}{S_{B1}} = \frac{2.077.378}{79.639} = 26,1 \text{ mm}$$

Element B2:

$$V_{B2} = \frac{\pi}{4} (D_4^2 - D_1^2) \cdot L_1 = \frac{\pi}{4} (230^2 - 50^2) \cdot 30 = 1.187.522 \text{ mm}^3$$

$$S_{B2} = \pi D_4 \cdot L_1 + \pi D_1 \cdot L_1 + \frac{\pi}{4} (D_4^2 - D_3^2) + \frac{\pi}{4} (D_2^2 - D_1^2)$$

$$S_{B2} = \pi \cdot 30 \cdot (230 + 50) + \frac{\pi}{4} (230^2 - 200^2) + \frac{\pi}{4} (80^2 - 50^2) = 39.584 \text{ mm}^2$$

$$M_{B2} = \frac{V_{B2}}{S_{B2}} = \frac{1.187.522}{39.584} = 30 \text{ mm}$$

Element B3:

$$V_{B3} = \frac{\pi}{4} (D_3^2 - D_2^2) \cdot L_2 = \frac{\pi}{4} (200^2 - 80^2) \cdot 120 = 3.166.725 \text{ mm}^3$$

$$S_{B3} = \pi D_3 \cdot L_2 + \pi D_2 \cdot L_2 + \frac{\pi}{4} (D_3^2 - D_2^2) = \pi \cdot 120 \cdot (200 + 80) + \frac{\pi}{4} (200^2 - 80^2) = 131.947 \text{ mm}^2$$

$$M_{B3} = \frac{V_{B3}}{S_{B3}} = \frac{3.166.725}{131.947} = 24 \text{ mm}$$

So $M_{B2} > M_{B1}$ and $M_{B2} > M_{B3}$, with

$$\frac{M_{B2}}{M_{B1}} = \frac{30}{26,1} = 1,15 \in [1,1 \div 1,3]$$

$$\frac{M_{B2}}{M_{B3}} = \frac{30}{24} = 1,25 \in [1,1 \div 1,3]$$

Therefore, the solidification will start in geometries B1 and B3, and will end in geometry B2.

A riser should be positioned in contact with the geometry B2.

- 3) *Considering: the axis of the casting as parting line, a volume of the casting cavity equal to 6,5 dm³, a riser with a volume equal to 3 dm³, a middle gating system with H_{av} equal to 100 mm and $c = 0,6$. Determine the area of the bottleneck so that the mould filling time is equal to 10 s.*

The pouring velocity in the bottleneck cross section (A_{bn}) will be

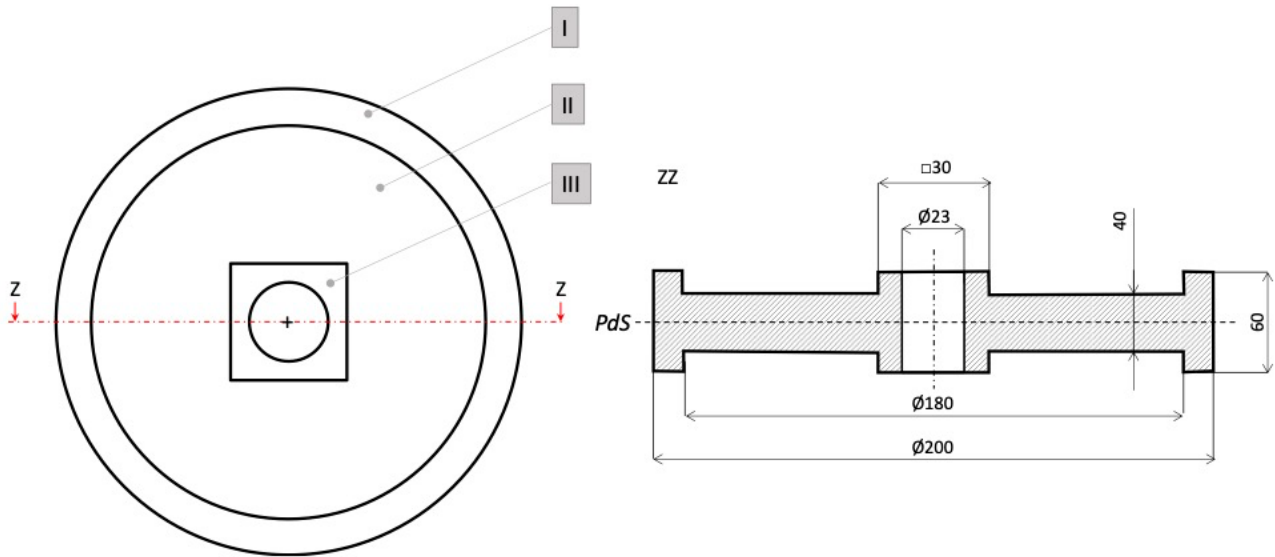
$$v = c \cdot \sqrt{2 \cdot g \cdot H_{av}} = 0,6 \cdot \sqrt{2 \cdot 9,81 \cdot \frac{100}{1000}} = 0,840 \frac{\text{m}}{\text{s}} = 840 \frac{\text{mm}}{\text{s}}$$

Being the mould filling time equal to $T_{MF} = \frac{V_{total}}{A_{bn} \cdot v}$, we get:

$$V_{total} = V_{casting} + V_{risers} = 6,5 + 3 = 9,5 \text{ dm}^3 = 9.500.000 \text{ mm}^3$$

$$A_{bottleneck} = \frac{V_{total}}{T_{MF} \cdot v} = \frac{9.500.000}{10 \cdot 840} = 1.131 \text{ mm}^2$$

Problem 4 (12/07/2021)



A steel lift pulley is obtained by sand casting using a middle gating system. The top view and the Z-Z cross-section of the main cavity in the mold are shown in the figure, where PdS is indicating the parting line. In the figure the decomposition into elementary geometries is also indicated (draft angles and fillet radii are neglected).

- 1) Study the directional solidification of the casting.
- 2) Considering a volume of the entire casting equal to $1,25 \cdot 10^6 \text{ mm}^3$, a total volume of open risers equal to 400 cm^3 , an initial pouring height equal to 100 mm, a friction factor $c = 0,6$, and a gating system ratio $A_s:A_r:A_g = 4:3:2$, design the gating system to ensure a mold filling time equal to 10 s.

Solution

- 1) Study the directional solidification of the casting.

Element I:

$$V_I = \frac{\pi}{4} (200^2 - 180^2) \cdot 60 = 358.142 \text{ mm}^3$$

$$A_I = 2 \cdot \frac{\pi}{4} (200^2 - 180^2) + \pi \cdot 200 \cdot 60 + \pi \cdot 180 \cdot (60 - 40) = 60.947 \text{ mm}^2$$

$$M_I = \frac{V_I}{A_I} = \frac{358.142}{60.947} = 5,9 \text{ mm}$$

Element II:

$$V_{II} = \frac{\pi}{4} 180^2 \cdot 40 - 30^2 \cdot 40 = \left(\frac{\pi}{4} 180^2 - 30^2 \right) \cdot 40 = 981.876 \text{ mm}^3$$

$$A_{II} = 2 \cdot \left(\frac{\pi}{4} 180^2 - 30^2 \right) = 49.094 \text{ mm}^2$$

$$M_{II} = \frac{V_{II}}{A_{II}} = \frac{981.876}{49.094} = 20 \text{ mm}$$

Element III:

$$V_{III} = 30^2 \cdot 60 - \frac{\pi}{4} 23^2 \cdot 60 = \left(30^2 - \frac{\pi}{4} 23^2 \right) \cdot 60 = 29.071 \text{ mm}^3$$

$$A_{III} = 2 \cdot \left(30^2 - \frac{\pi}{4} 23^2 \right) + 4 \cdot 30 \cdot 20 + \pi \cdot 23 \cdot 60 = 7.704 \text{ mm}^2$$

$$M_{III} = \frac{V_{III}}{A_{III}} = \frac{29.071}{7.704} = 3,8 \text{ mm}$$

Directional solidification: The first element that will solidify is Element III, then Element I and at last Element II. Being M_{II} too high with respect both M_I and M_{III} even if the directional solidification will be guaranteed, fracture may occur during the cooling at solid state.

- 2) Considering a volume of the entire cavity equal to $1,25 \cdot 10^6 \text{ mm}^3$, a total volume of open risers equal to 400 cm^3 , an initial pouring height equal to 100 mm , a friction factor $c = 0,6$, and a gating system ratio $A_s : A_r : A_g = 4 : 3 : 2$, design the gating system to ensure a mould filling time equal to 10 s .

Being the mold filling time equal to $T_{MF} = \frac{V_{total}}{A_{bn} \cdot v}$, we know that the minimal bottleneck cross section (A_{bn}), that is the area of the gates in this situation, will be $A_{bn} = A_g = \frac{V_{total}}{T_{MF} \cdot v}$.

Therefore:

$$V_{total} = V_{casting} + V_{risers} = 1.250.000 + 400.000 = 1.650.000 \text{ mm}^3$$

The pouring velocity in the bottleneck cross section will be $v = c \cdot \sqrt{2 \cdot g \cdot H_{av}}$, where:

$$H_{av} = \left(\frac{1}{\frac{r_{drag}}{\sqrt{h_i}} + \frac{r_{cope}}{\frac{\sqrt{h_i} + \sqrt{h_f}}{2}}} \right)^2$$

Being $h_i = 100 \text{ mm}$ and $h_f = 0 \text{ mm}$ (we are considering open risers), we get:

$$r_{drag} = \frac{V_{casting}/2}{V_{total}} = \frac{1.250.000/2}{1.650.000} = 0,38$$

$$r_{cope} = 1 - r_{drag} = 1 - 0,38 = 0,62$$

$$H_{av} = \left(\frac{1}{\frac{r_{drag}}{\sqrt{h_i}} + \frac{r_{cope}}{\frac{\sqrt{h_i} + \sqrt{h_f}}{2}}} \right)^2 = \frac{1}{\left(\frac{0,38}{\sqrt{100}} + \frac{0,62}{\frac{\sqrt{100} + \sqrt{0}}{2}} \right)^2} = 38,1 \text{ mm}$$

$$v = c \cdot \sqrt{2 \cdot g \cdot H_{av}} = 0,6 \cdot \sqrt{2 \cdot 9,81 \cdot \frac{38,1}{1000}} = 0,519 \frac{\text{m}}{\text{s}} < 1 \frac{\text{m}}{\text{s}}$$

Therefore:

$$A_g = \frac{V_{total}}{T_{MF} \cdot v} = \frac{1.650.000}{10 \cdot 0,519 \cdot 1000} = 318 \text{ mm}^2$$

$$A_r = \frac{3}{2} A_g = \frac{3}{2} \cdot 318 = 477 \text{ mm}^2$$

$$A_s = \frac{4}{2} A_g = 2 \cdot 318 = 636 \text{ mm}^2$$